CLOPIDOGREL

Bleeding risk

Prodrug (CYP2C19)

CHEMICAL NAME Methyl (S)-(2-chlorophenyl)(6,7-dihydrothieno[3,2-c]pyridin-5(4H)-yl)acetate (hydrogen sulfate)

CLASS P2Y12 ADP-receptor antagonist (thienopyridine antiplatelet)

TRADE NAMES Plavix · Iscover · Clopilet · Plagril

THERAPEUTIC USES / INDICATIONS

- Acute coronary syndrome and after coronary stenting (with aspirin; dual antiplatelet therapy).
- Secondary prevention after ischaemic stroke, MI or peripheral arterial disease.
- Aspirin intolerance.
- Minor stroke/high-risk TIA (short-term with aspirin).

ADMINISTRATION

- **PO** once daily with or without food.
- Do not stop before stent course completed unless cardiologist agrees.

DOSAGES

- **ACS / PCI loading:** 300–600 mg, then 75 mg once daily.
- **Maintenance:** 75 mg once daily.
- **DAPT duration:** usually 12 months after ACS (6 months after elective stent), per cardiology.
- **Stroke/TIA (with aspirin):** 300 mg load, then 75 mg daily for 21 days, then monotherapy.

PHARMACOKINETICS

- **Onset:** 2–6 h after loading; full effect 3–7 days with maintenance. **Duration:** 5–7 days (platelet lifetime).
- Prodrug activated by CYP2C19 in liver (poor metabolisers — reduced effect).
- Excreted in urine and faeces. **t½:** 6 h (active metabolite short).

NURSING MANAGEMENT

- Monitor for bleeding: stools, urine, gums, bruising, puncture sites.
- Check CBC (platelets, Hb).
- Use pantoprazole rather than omeprazole if PPI required.
- Make sure patient understands not to stop early after stent.
- Pre-procedure: confirm plan with prescriber.

PHARMACODYNAMICS

- Active metabolite irreversibly blocks P2Y12 ADP receptors on platelets → ↓ GP IIb/IIIa activation → ↓ aggregation.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Bruising, minor bleeding.
- Dyspepsia, diarrhoea.
- Rash.

ADVERSE EFFECTS (SERIOUS)

- Major bleeding (GI, intracranial).
- Thrombotic thrombocytopenic purpura (rare).
- Neutropenia; hypersensitivity.

CONTRAINDICATIONS & CAUTIONS

- Active pathological bleeding (peptic ulcer, intracranial haemorrhage).
- Severe hepatic impairment.
- **Caution:** surgery (stop 5–7 days before if advised), bleeding risk.

KEY INTERACTIONS

- **Omeprazole, esomeprazole:** ↓ activation — use pantoprazole if PPI needed.
- **Warfarin, DOACs, NSAIDs, SSRIs:** ↑ bleeding.
- **Fluconazole, fluoxetine (CYP2C19 inhibitors):** ↓ effect.

PATIENT EDUCATION

- Never stop without asking your cardiologist; risk of stent blockage and heart attack.
- Report black stools, blood in urine, severe headache.
- Tell dentists and surgeons you take it.
- Avoid NSAIDs.

KEY POINTS

- P2Y12 inhibitor; prodrug via CYP2C19.
- Avoid omeprazole — use pantoprazole.
- Load 300–600 mg, then 75 mg daily.
- Do not stop early after stents.